SPARKY'S SCIENCE EXPERIMENTS

Floating Egg: Salt Water and Density

Physical Science (floating and sinking) · Ages 5+ · 15 minutes

Make an egg float by adding salt to water, and find out why it is easier to float in the sea.

Safety first

Sparky always says: Wash your hands after touching raw eggs!

- An adult can help so the egg doesn't break.

You need

- A raw egg
- A tall clear glass or jar
- Tap water
- Table salt and a tablespoon

Steps

1. Fill the glass two-thirds with water. Gently put the egg in. Does it sink or float?
2. Take the egg out, add a tablespoon of salt and stir until it disappears.
3. Put the egg back. Keep adding salt, a spoon at a time, until the egg floats. Count the spoons.
4. Challenge: slowly pour plain water down the side of the glass, on top of the salty water. Can you make the egg float in the middle?

If it doesn't work, try these in order

1. Won't float? Keep adding salt and stir until it disappears - you may need 6 or more spoons.
2. Salt won't dissolve? Stir longer, or ask an adult for warm water.

Why it works

Density is how much stuff is packed into a space. An egg is a bit denser than tap water, so it sinks. Dissolving salt makes the water denser. Once the salty water is denser than the egg, the egg floats. In the challenge, plain water stays on top of the heavier salty water, so the egg floats in between.

Did you know? Sea water is salty, so swimmers and ships float a little higher in the sea at Durban or Cape Town than in a dam. A very old egg floats even in tap water, because air gets into it.

Ask Sparky: Would sugar work instead of salt? Is it easier to float in the sea or a pool?